2-Morita Equivalent Condensable Algebras in Topological Orders

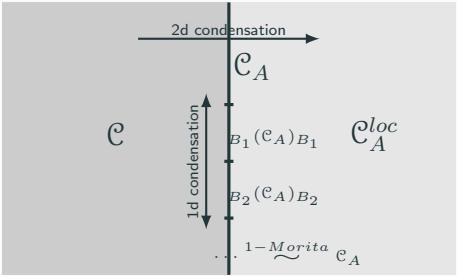
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PiTP 2024 Gong Show

Based on [arXiv: 2403.19779], a joint work with Holiverse Yang

Primer [Kitaev06, Bais-Slingerland-Haaker09, Davydov10, Kitaev-Kong12, Davydov-Müger-Nikshych-Ostrik13, Kong14]

- A 2d anomaly-free topological order can be described by a unitary modular tensor category (MTC) $\mathcal{C}$ (with a central charge c).
- A 2d condensable (separable commutative Frobenius) algebra $A \in \mathcal{C}$ can be condensed to create new topological order, described by the category $\mathcal{C}_A^{loc}$.
- Anyons that are confined from going to the condensed phase form a 1d domain wall described by a fusion category $\mathcal{C}_A$.
- A 1d condensable algebra $B \in \mathcal{C}_A$ can be condensed to create another domain wall ${}_B(\mathcal{C}_A)_B$.

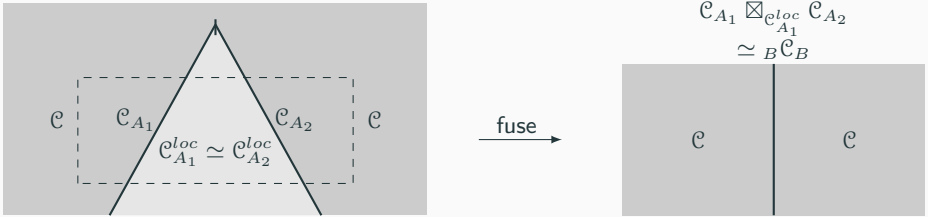


$$\mathcal{C}_{A_1}^{loc} \simeq \mathcal{C}_{A_2}^{loc}$$
$$\Updownarrow$$
$$A_1 \overset{2-\widetilde{Morita}}{} A_2$$

Motivation: classify 2-Morita equivalent A_i

Result I:

By taking left and right centers of 1d condensable algebras $B_i \in \mathcal{C}$, we can exhaust all 2-Morita equivalent 2d condensable algebras $\{A_i\}$ in $\mathcal{C}$.



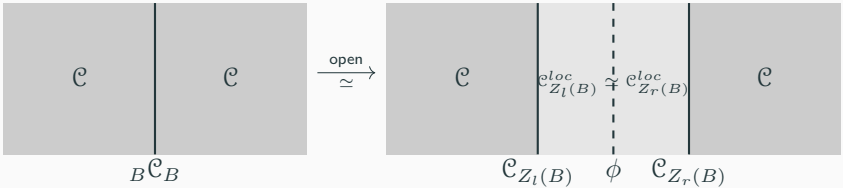
$$A_1 \simeq Z_l(B) \overset{2-Morita}{\sim} A_2 \simeq Z_r(B)$$

Theorem (Based on [Fröhlich-Fuchs-Runkel-Schweigert 06])

The pairs of 2-Morita equivalent condensable algebras in $\mathcal{C}$ are classified by left and right centers of $B_i \in \mathcal{C}$, in which the resulted category of local modules $\mathcal{C}_{Z_r(B)}^{loc}$ and $\mathcal{C}_{Z_r(B)}^{loc}$ are equivalent as modular tensor categories.

Result II:

Moreover, we proved that any any stable gapped domain wall in a 2d topological order can be 'pulled open'.

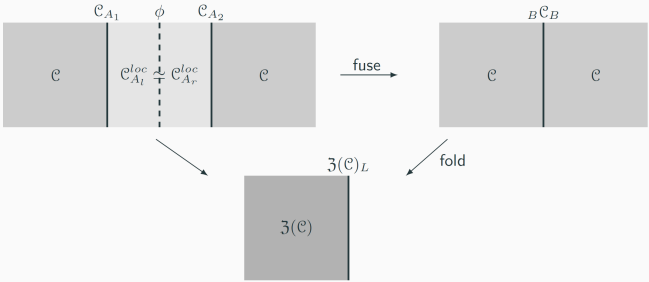


Theorem (Based on [Davydov-Nikshych-Ostrik 12])

An indecomposable $\mathcal{C}$-$\mathcal{C}$-bimodule in $\mathcal{C}$ can be written as ${}_B\mathcal{C}_B$ for some 1d condensable algebra $B \in \mathcal{C}$. We have ${}_B\mathcal{C}_B \simeq \mathcal{C}_{Z_l(B)} \boxtimes_{\mathcal{C}^{loc}_{Z_l(B)}} \phi \boxtimes_{\mathcal{C}^{loc}_{Z_r(B)}} \mathcal{C}_{Z_r(B)}$ where $\phi : \mathcal{C}^{loc}_{Z_l(B)} \xrightarrow{\sim} \mathcal{C}^{loc}_{Z_r(B)}$ is an equivalence of MTCs.

Result III:

The pairs of 2-Morita equivalent condensable algebras in $\mathcal{C}$ can also be classified by lagrangian algebras (maximal 2d condensable algebras) L_i's in $\mathcal{C} \boxtimes \overline{\mathcal{C}} \simeq \mathfrak{Z}(\mathcal{C})$.



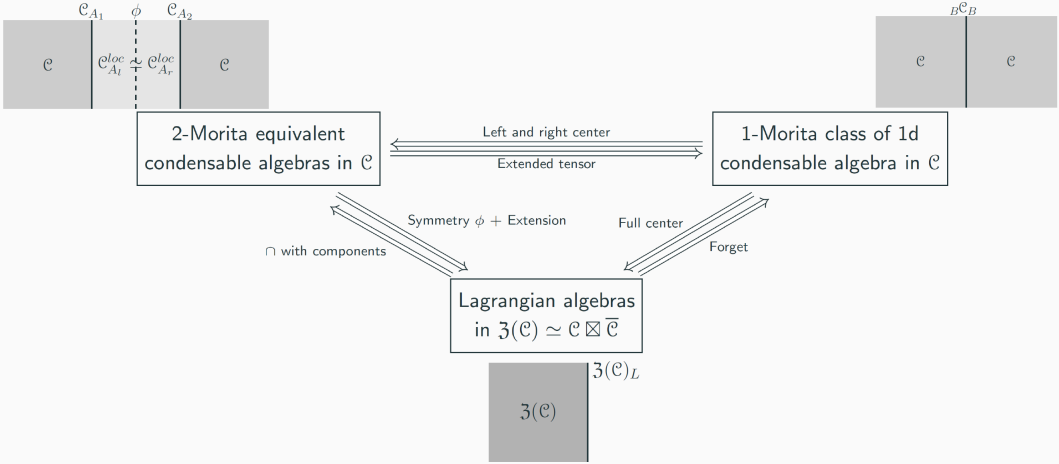
$$L_i \cap (\mathcal{C} \boxtimes \mathbf{1}) =: A_{l_i} \overset{2-Morita}{\sim} A_{r_i} := L_i \cap (\mathbf{1} \boxtimes \overline{\mathcal{C}})$$

Theorem

Given any pair of 2-Morita equivalent algebras (A_l, A_r) *in* $\mathcal{C}$, $\mathcal{C}_{A_l} \boxtimes_{\mathcal{C}^{loc}_{A_l}} \phi \boxtimes_{\mathcal{C}^{loc}_{A_r}} \mathcal{C}_{A_r}$ *is equivalent to* $\mathfrak{Z}(\mathcal{C})_L (\simeq \mathfrak{Z}(\mathcal{C})_{Z(B)}$ *[Kong-Runkel 09]) as monoidal* $\mathcal{C}$-$\mathcal{C}$*-bimodule for some lagrangian algebra* $L \in \mathfrak{Z}(\mathcal{C})$.

Main results:

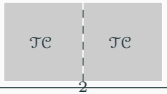
We give a complete interplay between E_1 condensable algebras in $\mathcal{C}$, 2-Morita equivalent E_2 condensable algebras in $\mathcal{C}$, and lagrangian algebras in $\mathcal{C} \boxtimes \overline{\mathcal{C}}$.



Example:

2d Toric code model $\mathcal{TC} := \mathfrak{Z}(\mathrm{Rep}(\mathbb{Z}_2))$ $\mathsf{H} := \sum_v (1 - \mathsf{A}_v) + \sum_p (1 - \mathsf{B}_p)$

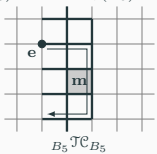
$B_i \in \mathcal{TC}$	$Z_l(B_i)/Z_r(B_i)$	Domain wall	Lagrangian algebras $L_i \in \mathcal{TC} \boxtimes \overline{\mathcal{TC}}$
$\mathbf{1}$	$\mathbf{1}/\mathbf{1}$	trivial wall	$\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{e}\bar{\mathbf{e}} \oplus \mathbf{m}\bar{\mathbf{m}} \oplus \mathbf{f}\bar{\mathbf{f}}$
$\mathbf{1} \oplus \mathbf{f}$	$\mathbf{1}/\mathbf{1}$	$\mathbf{e} - \mathbf{m}$ exchange	$\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{m}\bar{\mathbf{e}} \oplus \mathbf{e}\bar{\mathbf{m}} \oplus \mathbf{f}\bar{\mathbf{f}}$
$\mathbf{1} \oplus \mathbf{e}$	$\mathbf{1} \oplus \mathbf{e}/\mathbf{1} \oplus \mathbf{e}$	$\mathrm{Vec}_{\mathbb{Z}_2} \boxtimes \mathrm{Vec}_{\mathbb{Z}_2}$	$\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{e}\bar{\mathbf{1}} \oplus \mathbf{1}\bar{\mathbf{e}} \oplus \mathbf{e}\bar{\mathbf{e}}$
$\mathbf{1} \oplus \mathbf{m}$	$\mathbf{1} \oplus \mathbf{m}/\mathbf{1} \oplus \mathbf{m}$	$\mathrm{Rep}(\mathbb{Z}_2) \boxtimes \mathrm{Rep}(\mathbb{Z}_2)$	$\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{m}\bar{\mathbf{1}} \oplus \mathbf{1}\bar{\mathbf{m}} \oplus \mathbf{m}\bar{\mathbf{m}}$
$\mathbf{1} \oplus \mathbf{e} \oplus \mathbf{m} \oplus \mathbf{f}$	$\mathbf{1} \oplus \mathbf{e}/\mathbf{1} \oplus \mathbf{m}$	$\mathrm{Vec}_{\mathbb{Z}_2} \boxtimes \mathrm{Rep}(\mathbb{Z}_2)$	$\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{e}\bar{\mathbf{1}} \oplus \mathbf{1}\bar{\mathbf{m}} \oplus \mathbf{e}\bar{\mathbf{m}}$
$\mathbf{1} \oplus \mathbf{e} \oplus \mathbf{m} \oplus \mathbf{f}, \omega$	$\mathbf{1} \oplus \mathbf{m}/\mathbf{1} \oplus \mathbf{e}$	$\mathrm{Rep}(\mathbb{Z}_2) \boxtimes \mathrm{Vec}_{\mathbb{Z}_2}$	$\mathbf{1}\bar{\mathbf{1}} \oplus \mathbf{m}\bar{\mathbf{1}} \oplus \mathbf{1}\bar{\mathbf{e}} \oplus \mathbf{m}\bar{\mathbf{e}}$

H	F	E_2 condensable algebras in $\mathcal{TC}$	Condensed phase $\mathcal{TC}_A^{loc}$	Domain walls	Total: 6
$\mathbb{Z}_2$ $\{e\}$	$\mathbb{Z}_2$ $\{e\}$	$\mathbf{1} \oplus \mathbf{m}$ $\mathbf{1} \oplus \mathbf{e}$	Vec		non-invertible: $2 \times 2 = 4$
$\mathbb{Z}_2$	$\{e\}$	$\mathbf{1}$	$\mathcal{TC}$		invertible: 2

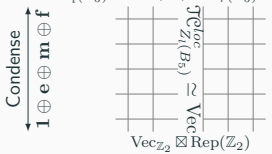
We can actually realize 1d condensable algebras together with their left/right centers in the lattice model:



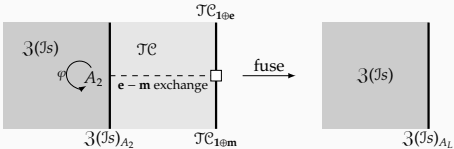
$Z_l(B_5) \simeq \mathbf{1} \oplus \mathbf{e}$ $Z_r(B_5) \simeq \mathbf{1} \oplus \mathbf{m}$



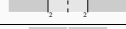
$\mathcal{TC}_{Z_l(B_5)} \simeq \mathrm{Vec}(\mathbb{Z}_2)$ $\mathcal{TC}_{Z_r(B_5)} \simeq \mathrm{Rep}(\mathbb{Z}_2)$



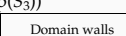
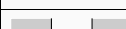
$$\mathsf{H}_{wall} = \mathsf{H} + \sum_k \mathsf{A}_{v_{0,k}} + \sum_k \mathsf{B}_{p_{\frac{1}{2},k-\frac{1}{2}}} = \sum_{v \neq v_{0,k}} (1 - \mathsf{A}_v) + \sum_{p \neq p_{\frac{1}{2},k-\frac{1}{2}}} (1 - \mathsf{B}_p) + 2N$$



Results of $\mathfrak{Z}(\mathrm{Rep}(\mathbb{Z}_4))$

H	F	Condensable algebras in $\mathfrak{Z}(\mathrm{Rep}(\mathbb{Z}_4))$	Condensed phase $\mathfrak{Z}(\mathrm{Rep}(\mathbb{Z}_4))^{loc}_A$	Domain walls	Total: 22
$\{e\}$	$\{e\}$	$\mathbf{1} \oplus \mathbf{e} \oplus \mathbf{e}^2 \oplus \mathbf{e}^3$	Vec		non-invertible: $3 \times 3 = 9$
$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbf{1} \oplus \mathbf{e}^2 \oplus \mathbf{m}^2 \oplus \mathbf{f}^2$			
$\mathbb{Z}_4$	$\mathbb{Z}_4$	$\mathbf{1} \oplus \mathbf{m} \oplus \mathbf{m}^2 \oplus \mathbf{m}^3$			
$\mathbb{Z}_4$	$\mathbb{Z}_2$	$\mathbf{1} \oplus \mathbf{f}^2$	$\mathcal{DS}$		non-invertible: 1
$\mathbb{Z}_4$	$\mathbb{Z}_2$	$\mathbf{1} \oplus \mathbf{m}^2$	$\mathcal{TC}$		non-invertible: $2 \times 2 \times 2 = 8$
$\mathbb{Z}_2$	$\{e\}$	$\mathbf{1} \oplus \mathbf{e}^2$			
$\mathbb{Z}_4$	$\{e\}$	$\mathbf{1}$	$\mathfrak{Z}(\mathrm{Rep}(\mathbb{Z}_4))$		invertible: 4

Results of $\mathfrak{Z}(\mathrm{Rep}(S_3))$

H	F	Condensable algebras in $\mathfrak{Z}(\mathrm{Rep}(S_3))$	Condensed phase $\mathfrak{Z}(\mathrm{Rep}(S_3))^{loc}_A$	Domain walls	Total: 28
S_3	S_3	$\mathbf{A} \oplus \mathbf{F} \oplus \mathbf{D}$	Vec		non-invertible: $4 \times 4 = 16$
A_3	A_3	$\mathbf{A} \oplus \mathbf{B} \oplus \mathbf{2F}$			
C_2	C_2	$\mathbf{A} \oplus \mathbf{C} \oplus \mathbf{D}$			
$\{e\}$	$\{e\}$	$\mathbf{A} \oplus \mathbf{B} \oplus \mathbf{2C}$			
S_3	A_3	$\mathbf{A} \oplus \mathbf{F}$	$\mathcal{TC}$		non-invertible: $2 \times 2 \times 2 = 8$
C_2	$\{e\}$	$\mathbf{A} \oplus \mathbf{C}$			
A_3	$\{e\}$	$\mathbf{A} \oplus \mathbf{B}$	$\mathfrak{Z}(\mathrm{Rep}(\mathbb{Z}_3))$		non-invertible: $4/2 = 2$
S_3	$\{e\}$	$\mathbf{A}$	$\mathfrak{Z}(\mathrm{Rep}(S_3))$		invertible: 2